

MA.6.GR.1.2

Benchmark

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.1.2, MA.6.NSO.1.3, MA.6.NSO.1.4
- MA.6.NSO.4.1
- MA.6.AR.2.4

Terms from the K-12 Glossary

- Axes (of a graph)
- Coordinate
- Coordinate Plane
- Origin
- Quadrant

Vertical Alignment**Previous Benchmarks**

- MA.5.GR.4.2

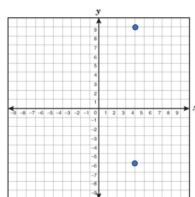
Next Benchmarks

- MA.8.GR.1.2

Purpose and Instructional Strategies

Throughout the elementary grades, students have used the number line to model subtraction as the distance between two points. In grade 6, this connection has been extended to negative integers, and in this benchmark, it extends to the distance between ordered pairs of rational numbers in the four quadrants, if they share a coordinate. This prepares students in a visual way for the subtraction of rational numbers in grade 7. In grade 8, students will find distances between any two ordered pairs using the Pythagorean Theorem.

- Instruction connects student understanding of MA.6.NSO.1 to the coordinate plane. Strategies include using absolute value to find the distance between two ordered pairs. Even though in grade 6, students are adding and subtracting integers, this benchmark focuses on using absolute value to explore addition and subtraction of rational numbers.
 - For example, the points $(4, 9)$ and $(4, -6)$ are plotted on a coordinate grid.



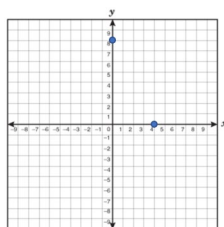
Students can use the absolute value of the first y -coordinate, 9, and the second y -coordinate, -6 , and add these numbers together to determine the distance is 15.

- When using rational numbers, instruction is restricted to numbers within the same form.
 - For example, if students are working with fractions, the ordered pairs will not include decimals. If students are working with decimals, the ordered pairs will not

include fractions. Students should not be penalized if they convert from one form to another when performing operations. Students should be given the opportunity to find the distance between the ordered pairs both on and off of a graph.

Common Misconceptions or Errors

- Students may have trouble finding the distance across either of the axes.
 - For example, find the distance between $(-4, 3)$ and $(5, 3)$. It may appear that the “difference” between -4 and 5 is 1 . However, the distance from -4 to 0 is 4 units and from 0 to 5 is 5 units. Therefore, the distance between the two points is 9 units.
- Students may misunderstand that a point on an axis has at least one coordinate of zero.
 - For example, the points $(0, 8)$ and $(4, 0)$ are graphed on the coordinate plane.
 - Some students may incorrectly believe that the points represent $(8, 8)$ and $(4, 4)$ or just (8) and (4) .



Strategies to Support Tiered Instruction

- Instruction includes using two notecards and covering empty portions of the coordinate plane to focus attention on the space between two provided points. Students can then count the number of spaces between the two points, paying attention to scaling.
- If points are not already placed on a coordinate plane, students may plot the points to create a visual representation of the distances created.
- Instruction includes building connections to plotting points on number lines. On one sheet of tracing paper, label a horizontal number line as the x -axis and plot the x -value. On another sheet of tracing paper, label a vertical number line as the y -axis and plot the y -value. Overlap the two number lines with a point of intersection at the origin $(0,0)$. Place a third sheet of tracing paper on top of the two number lines, trace and label the x - and y -axis to produce a coordinate plane. The location of a new point should then be plotted in the appropriate quadrant to represent the horizontal and vertical locations of the two previously plotted points. This same strategy helps to draw connections to a point laying on an axis if one of the coordinates is zero.
- Instruction includes creating connections back to finding distance on a number line. Lay a piece of tracing paper on top of the provided coordinate plane, trace the points, and draw a number line through the two points, paying close attention to the scaling. Once the number line is drawn, remove the tracing paper and find the distance between the two points on the number line.

Instructional Tasks

Instructional Task 1 (MTR.6.1)

On a coordinate plane, Perry is planning a future town by labeling different buildings. Each unit on the coordinate plane is equivalent to 0.5 miles.

- Town Hall (2, 5)
- Fire Station (-4, 5)
- Library (-4, -7)
- Ice Cream Shop (3, -5)
- Central Park (0, 0)
- Grocery Store (2, -8)
- Police Station (0, 6)
- Gas Station (-4, -5)

Part A. What is the distance, in miles, from the Fire Station to the Library?

Part B. How far, in miles, would the mayor, located at Town Hall, have to walk to get to the grocery store?

Part C. Compare a walk from Central Park to the Police Station and a walk from the Gas Station to the Ice Cream Shop. Which one is further?

Part D. Using what you have learned can you determine how far Perry would have to ride his bike, in miles, if he is starting at Central Park and going to the Ice Cream Shop and can only travel North, South, East or West.

Instructional Task 2 (MTR.3.1)

André is located at (-5, 3), Boris is located at (-1, 3) and Carlos is located at (-5, -2). Determine the distance between André and Boris. Then determine the distance between André and Carlos. Show your work.

Instructional Items

Instructional Item 1

The points (0.4, -6) and (1.9, -6) represent the location of two towns on a coordinate plane, where one unit is equal to one mile. What is the distance, in miles, between the two towns?

Instructional Item 2

If the distance between two points is 7 units and one of the points is located at (-6, 3), what could be the coordinates of the other point?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*